
UNDERSTANDING THE FINE-GRAINED KNOWLEDGE CAPABILITIES OF VISION-LANGUAGE MODELS

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ABSTRACT

Vision-language models (VLMs) have attained substantial gains on visual question answering benchmarks that probe visual reasoning, document understanding, and multimodal dialogue. These gains are consistent across VLMs instantiated with heterogeneous base language models, alignment architectures, and training corpora. Nevertheless, recent studies report that such models underperform on conventional image classification benchmarks, which measure fine-grained visual knowledge. We conduct a systematic evaluation of a large set of recent VLMs on fine-grained classification benchmarks and isolate the factors underlying the discrepancy between fine-grained knowledge and performance on other vision benchmarks. Via a controlled series of ablation experiments, we establish that substituting a stronger base language model yields uniform improvements across all benchmark scores, whereas substituting a stronger vision encoder yields disproportionately larger improvements on fine-grained classification. We further establish that the pretraining stage is decisive for fine-grained performance, particularly under the configuration in which the language model weights are unfrozen during pretraining. These findings delineate concrete design levers for improving fine-grained visual understanding and vision-centric capabilities in VLMs.

1 INTRODUCTION

Vision-language models (VLMs) couple a visual encoder with a large language model (LLM) and have exhibited progressively stronger capabilities in interpreting and reasoning over visual content (Alayrac et al., 2022; Team et al., 2023; OpenAI, 2023; Liu et al., 2023; Wang et al., 2024b). Instantiations of this architecture perform complex tasks including open-ended visual question answering, document understanding, and multimodal dialogue (Goyal et al., 2017; Yue et al., 2024; Mathew et al., 2021; OpenAI, 2023). Despite this progress, a central question remains unresolved: to what degree do VLMs succeed on fine-grained visual perception tasks, and which components and training procedures govern that performance?

Fine-grained visual recognition—the discrimination of visually similar categories—requires a model to resolve subtle discriminative features while remaining invariant to intra-class nuisance factors such as pose, illumination, and background. Conventional contrastive and self-supervised vision encoders, notably CLIP (Radford et al., 2021) and DINO (Caron et al., 2021), attain strong accuracy on fine-grained classification benchmarks, yet the corresponding behavior of more recent VLMs has been comparatively underexamined.

Characterizing the fine-grained knowledge of VLMs is essential because perception constitutes the substrate on which higher-level understanding and reasoning are built. Deployed VLMs frequently operate over long-tailed distributions in which the discrimination of closely related categories is decisive. As a concrete instance, a model that misidentifies a mushroom species cannot subsequently determine, from its textual knowledge, whether the specimen is toxic or safe for consumption. Furthermore, quantifying the factors that govern fine-grained knowledge acquisition can directly inform the architectural and training design of future VLMs.

This paper first reports a systematic evaluation of fine-grained visual knowledge in state-of-the-art VLMs, contrasting their fine-grained classification accuracy with their performance on general VLM tasks. Concretely, we evaluate 15 VLMs—including LLaVA (Liu et al., 2023; 2024a), Phi (Abdin

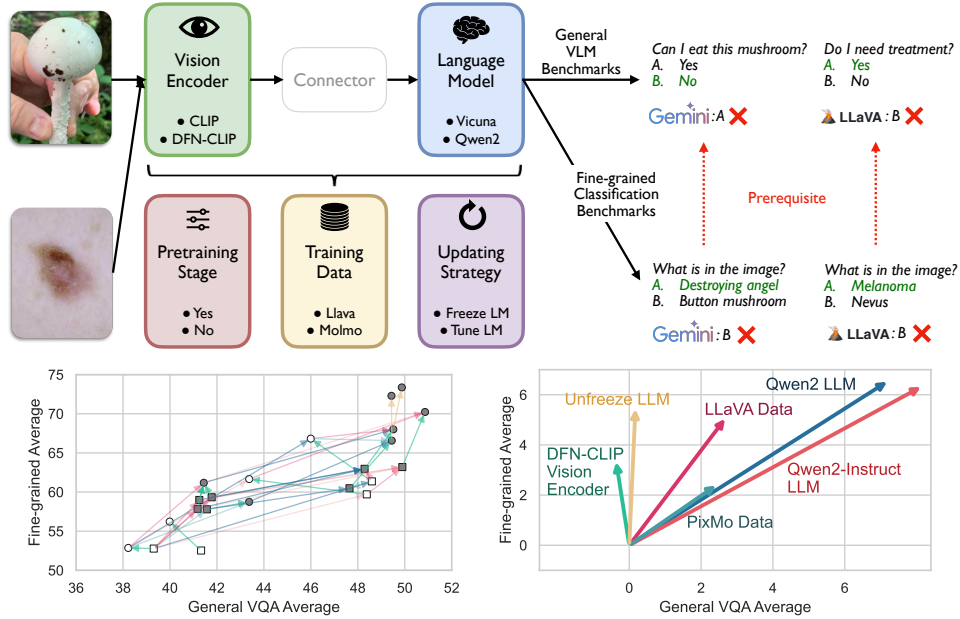


Figure 1: **Overview.** (Top) We characterize the fine-grained classification capabilities of vision-language models (VLMs)—a foundational competence that underpins higher-level understanding and reasoning yet is seldom measured directly. (Bottom) Across 22 controlled ablation experiments that vary individual model components and training procedures (left), we quantify the contribution of each factor to fine-grained classification performance in VLMs (right).

et al., 2024), Qwen2-VL (Wang et al., 2024b), and Molmo (Deitke et al., 2024)—on four fine-grained classification benchmarks (ImageNet (Deng et al., 2009), Flowers (Nilsback & Zisserman, 2008), Pets (Parkhi et al., 2012), and Food (Bossard et al., 2014)), and juxtapose these measurements with an aggregate score over eight general VLM benchmarks including MMMU (Yue et al., 2024), MathVista (Lu et al., 2024), and MMVet (Yu et al., 2024). The analysis reveals substantial variance in fine-grained classification accuracy among models of comparable general capability, indicating that fine-grained recognition constitutes a distinct axis of visual intelligence that existing VLM benchmarks do not measure. In addition, we observe a persistent accuracy gap between each VLM and its own CLIP-based vision encoder, consistent with prior reports (Zhang et al., 2024; Geigle et al., 2024).

To identify the determinants of fine-grained classification performance in VLMs, we conduct a controlled study that isolates the role of individual model components and training procedures. Specifically, we quantify the effect of architectural choices—the base language model and the vision encoder—and of training strategies—the presence of a pretraining stage and variation in data quality—on classification accuracy. Across 22 controlled ablation experiments, we identify the following factors as significant determinants of fine-grained knowledge acquisition in VLMs:

- **Language model (§4.2):** Substituting a stronger base language model improves performance approximately uniformly across both fine-grained classification and general VLM benchmarks.
- **Vision encoder (§4.2):** Substituting a stronger vision encoder improves fine-grained classification performance while exerting limited effect on general VLM benchmarks, an asymmetry that is most pronounced in models trained under the two-stage protocol.
- **Pretraining stage (§4.3):** Large-scale pretraining on image-caption corpora substantially increases fine-grained classification performance while producing a smaller increase on general VLM benchmarks.
- **Weight updating method (§4.3):** Jointly updating the vision-encoder-to-LLM connector and the LLM parameters during pretraining yields higher fine-grained classification accuracy than updating the connector alone.
- **Data quality (§4.3):** The quality of the pretraining and instruction-tuning corpora exerts limited effect on fine-grained classification, an effect that is further attenuated when the language model weights are held frozen during pretraining.

In summary, this study quantifies fine-grained visual perception in VLMs—a capability that current VLM benchmark suites do not directly measure. Through controlled ablations, we identify and rank the factors that govern fine-grained knowledge acquisition in VLMs. The resulting findings specify concrete design levers for constructing vision-centric VLMs with stronger visual understanding, thereby improving their reliability on real-world long-tailed tasks that require fine-grained recognition.

2 RELATED WORK

Vision-language models and their evaluation. Vision-language models (VLMs) constitute a class of models that condition a language model on visual inputs, and predominantly adopt the LLaVA-style architecture (Liu et al., 2023) composed of three modules: a vision encoder, a language model, and an MLP connector that projects visual features into the language-model embedding space. Such models attain strong performance across a range of multimodal tasks, including visual question answering (VQA), document understanding, and general reasoning (Liu et al., 2023; Dai et al., 2023; Awadalla et al., 2023; Wang et al., 2024b). A large number of benchmarks have been introduced to evaluate them, including MMMU (Yue et al., 2024), MathVista (Wang et al., 2024a), and DocVQA (Mathew et al., 2021). These benchmarks, however, predominantly probe reasoning and language understanding conditioned on visual inputs, and do not measure core vision-centric competencies such as object recognition and fine-grained classification. This work addresses that gap through a systematic evaluation and analysis of VLMs on fine-grained classification, an essential but underexamined component of visual intelligence.

Fine-grained visual classification. Fine-grained classification is a well-established computer vision task whose objective is to discriminate visually similar subcategories nested within broader object categories, such as bird species (Wah et al., 2011), flowers (Nilsback & Zisserman, 2008), and pet breeds (Parkhi et al., 2012). The corresponding benchmarks have served as principal testbeds for training and evaluating vision encoders, including ResNet (He et al., 2016), CLIP (Radford et al., 2021), and DINO (Caron et al., 2021). Notwithstanding their centrality to vision research, recent VLMs have largely omitted these datasets from their evaluation protocols in favor of visual question answering. Because fine-grained classification is a prerequisite for real-world applications that operate over long-tailed distributions and demand precise recognition (e.g., medical diagnosis, food safety, and species identification), evaluating VLMs on these benchmarks is necessary for characterizing their practical utility from a vision-centric rather than a language-centric standpoint.

Evaluation and ablation of VLMs for fine-grained classification. Prior studies that evaluate VLMs on fine-grained classification (Zhang et al., 2024; Geigle et al., 2024) are limited in scope, typically restricted to a small number of models or datasets. We extend this line of work by evaluating a substantially larger set of VLMs across multiple fine-grained classification benchmarks under a single unified multiple-choice protocol. Beyond evaluation, we contribute a controlled ablation analysis that isolates the factors governing fine-grained classification performance.

A parallel body of work characterizes the VLM design space, analyzing model architectures (McKinzie et al., 2024), training strategies (Karamcheti et al., 2024), and the contribution of multimodal datasets (Gadre et al., 2023; Udandarao et al., 2024; Deitke et al., 2024). These analyses, however, are conducted predominantly on general benchmarks that emphasize language and reasoning. In contrast, we quantify how specific design choices—base-model selection, the pretraining procedure, and the composition of the pretraining and instruction-tuning corpora—affect fine-grained classification, yielding design guidance for optimizing VLMs on vision-centric tasks.

3 EVALUATION: FINE-GRAINED CLASSIFICATION

This section states the motivation for evaluating fine-grained classification, specifies the benchmarks and evaluation protocol, and reports the principal observations obtained from evaluating 15 existing VLMs.

3.1 MOTIVATION

Fine-grained classification is a fundamental computer vision task whose objective is to discriminate visually similar subcategories nested within a broader category (e.g., distinct bird species or car models) (Wei et al., 2021). Numerous evaluation benchmarks, including ImageNet (Deng et al., 2009) and Flowers (Nilsback & Zisserman, 2008), have been established and widely adopted as standard testbeds for training and evaluating vision encoders. With the maturation of large-scale self-supervised and supervised learning, vision encoders such as CLIP (Radford et al., 2021) and DINO (Caron et al., 2021) have attained near-human or superhuman accuracy on demanding fine-grained classification tasks.

These benchmarks have nonetheless been omitted from the evaluation of vision-language models (VLMs). Although recent VLMs such as LLaVA (Liu et al., 2023) exhibit strong capabilities in interpreting visual concepts, their evaluation is typically confined to visual question answering (VQA)—for instance, academic VQA benchmarks such as MMMU (Yue et al., 2024) and Math-Vista (Lu et al., 2024), or document understanding tasks such as DocVQA (Mathew et al., 2021) and TextVQA (Singh et al., 2019). Because these benchmarks emphasize reasoning and language processing and contain no fine-grained object categories, they leave core vision-centric capabilities effectively unmeasured.

Fine-grained classification is a prerequisite for VLMs deployed in real-world settings, which frequently operate over long-tailed distributions that demand precise visual discrimination among similar categories. Consider, for example, a visually impaired user querying whether a mushroom is safe to eat: the model must first identify the species correctly before it can reason about toxicity, and misclassifying a lethal destroying angel as an edible white button mushroom carries fatal consequences. Analogously, in medical diagnosis, failure to discriminate between clinically similar conditions can lead to incorrect treatment and serious harm. In autonomous driving, conflating a “stop” sign with a “do not enter” sign—both red and of similar geometry—can induce dangerous navigation errors. Fine-grained classification competence is therefore a necessary condition for the safety and reliability of VLMs in real-world deployment.

3.2 BENCHMARKS

We measure the fine-grained visual knowledge of VLMs on four well-established object recognition benchmarks, each targeting a distinct domain of fine-grained classification: **ImageNet-1K** (Deng et al., 2009), **Oxford Flowers-102** (Nilsback & Zisserman, 2008), **Oxford-IIIT Pet-37** (Parkhi et al., 2012), and **Food-101** (Bossard et al., 2014).

These benchmarks span 37 to 1,000 distinct classes, which precludes direct evaluation of VLMs in the multiple-choice format that is standard for VLM training and evaluation. To reconcile this, we follow the protocol of Geigle et al. (2024) and recast each dataset as a 5-way multiple-choice task. Concretely, for each image we use an OpenCLIP ViT-L/14 model (Schuhmann et al., 2022) to retrieve the four hardest negative labels, ranked by image-label cosine similarity, and combine them with the ground-truth label to form the answer set. We deliberately select the OpenCLIP model as the negative-mining backbone so that difficulty is not biased against VLMs whose encoders are OpenAI-CLIP (Radford et al., 2021), SigLIP (Zhai et al., 2023), or DFN-CLIP (Fang et al., 2023), each of which is trained on a corpus disjoint from that of the OpenCLIP model. This reformulation permits evaluation of VLMs in their native input format while retaining most of the discriminative difficulty of the original task. As a calibration, CLIP ViT-L/14-336px (Radford et al., 2021) attains 75.2% top-1 accuracy on the 102-class Flowers-102, which rises only to 78.3% on our 5-way multiple-choice reformulation.

3.3 TESTING SETUP

We evaluate 15 recent VLMs in the 7B–13B parameter range on the reformulated 5-way fine-grained classification benchmarks (Liu et al., 2023; 2024a; Chen et al., 2024a; Abidin et al., 2024; Deitke et al., 2024; Laurençon et al., 2024; Hong et al., 2024; Chen et al., 2024c; Wang et al., 2024b). For each VLM we adopt its default prompt template from VLMEvalKit (Duan et al., 2024) and score accuracy by exact match against the multiple-choice options. As a reference baseline, we evaluate several CLIP models (Radford et al., 2021; Schuhmann et al., 2022; Fang et al., 2023)

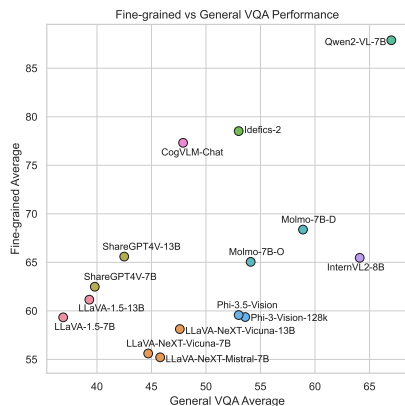


Figure 2: **Fine-grained classification versus general VQA performance across VLMs.** Across recent VLMs, fine-grained classification accuracy varies substantially at fixed general-VQA accuracy, indicating that it constitutes a distinct axis of visual capability not captured by standard VQA benchmarks.

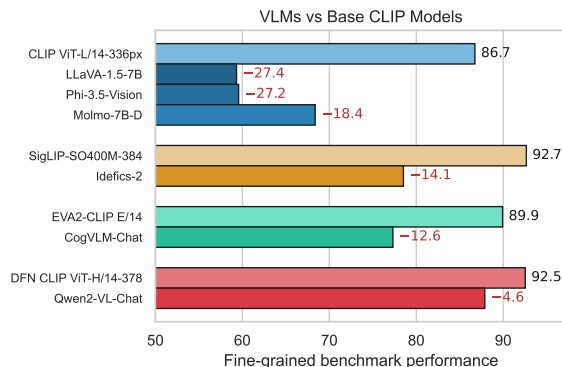


Figure 3: **Fine-grained classification accuracy of VLMs relative to their constituent CLIP vision encoders.** Qwen2-VL-Chat nearly recovers the accuracy of its DFN-CLIP encoder, whereas every other VLM trails its encoder by a wide margin, quantifying the headroom that remains for VLMs on fine-grained classification.

under the standard zero-shot protocol, in which each image and each class label (rendered as “A photo of a [class]”) are mapped to embeddings by CLIP’s image and text encoders, respectively. The top-1 prediction is the class whose text embedding maximizes the cosine similarity with the image embedding.

To determine whether existing VLM evaluation benchmarks capture fine-grained classification capability, we contrast each VLM’s fine-grained classification accuracy with its mean score over eight general VQA benchmarks from VLMEvalKit, including MMMU (Yue et al., 2024) and Math-Vista (Lu et al., 2024).

3.4 RESULTS

Figure 2 plots fine-grained classification accuracy (averaged over our four benchmarks) against general VQA accuracy (averaged over eight existing VQA benchmarks) for every evaluated VLM. At approximately fixed general-VQA accuracy, fine-grained classification accuracy varies markedly across models. For instance, CogVLM-Chat (Hong et al., 2024) and LLaVA-NeXT-Vicuna-13B (Liu et al., 2024a) attain nearly identical general-VQA accuracy (approximately 48%), yet their fine-grained classification accuracy differs by 19 percentage points (pp)—77.3% for CogVLM-Chat versus 58.1% for LLaVA-NeXT-Vicuna-13B. This dissociation demonstrates that existing VLM evaluation benchmarks do not capture fine-grained classification capability, establishing it as a distinct dimension of visual intelligence.

Finding 1: Fine-grained classification represents a distinct aspect of visual capability that current VLM benchmarks fail to adequately measure.

A structural property of current VLMs is that each is constructed on top of a pretrained vision encoder. Under the multiple-choice reformulation, this permits a direct, matched comparison between each VLM and its underlying CLIP encoder on traditional classification benchmarks. Figure 3 contrasts the fine-grained classification accuracy of selected VLMs with that of their constituent CLIP baselines, exposing a substantial gap. Molmo (Deitke et al., 2024), a recent model that performs strongly on several VQA benchmarks, attains only 68.4% mean accuracy under our multiple-choice protocol, whereas its CLIP ViT-L/14 encoder (Radford et al., 2021) attains 86.7%. Even the strongest evaluated VLM, Qwen2-VL-Chat (Wang et al., 2024b), trails its own state-of-the-art DFN-CLIP (Fang et al., 2023) encoder by 4.6 pp. This gap motivates a controlled analysis of VLM

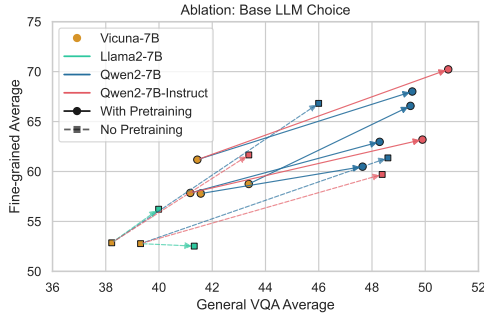


Figure 4: Base-LLM ablation: substituting Vicuna-7B in LLaVA with alternative LLMs. Averaged across configurations, replacing Vicuna with Qwen2-7B yields a +7.5pp increase in fine-grained accuracy and a +7.7pp increase in general VQA accuracy.

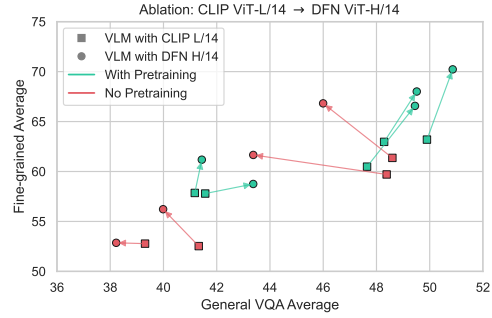


Figure 5: Base vision-encoder ablation: substituting CLIP ViT-L/14 in LLaVA with DFN-CLIP ViT-H/14. Conditioned on sufficient pretraining, substituting DFN-CLIP raises fine-grained accuracy by +4.5pp and general accuracy by +1.2pp.

design choices to identify the factors that improve fine-grained classification capability, which we undertake through the ablation study in the following section.

Finding 2: A significant performance gap exists between VLMs and CLIP models in fine-grained classification tasks.

4 ABLATIONS: BASE MODELS AND TRAINING

4.1 EXPERIMENTAL SETUP

Building on the observational results of Section 3, we conduct controlled ablations over the inter-model differences that may account for fine-grained classification performance, targeting primarily base-model selection and training methodology. All experiments are implemented within the LLaVA-1.5 training framework and alignment architecture (Liu et al., 2023). Each model is trained for one epoch of pretraining (where applicable) followed by one epoch of finetuning. To isolate the effect of the variable under study, we hold LLaVA-1.5’s default hyperparameters—including learning rate and batch size—fixed across all experiments.

Fine-grained performance is quantified as the mean accuracy over the four classification benchmarks specified in Section 3.2. General VQA performance is quantified as the mean score over three widely used multiple-choice benchmarks: MMMU (Yue et al., 2024), MMBench (Liu et al., 2024b), and MMStar (Chen et al., 2024b). These three were selected for their domain coverage and established adoption in VLM evaluation.

4.2 BASE MODELS

The analysis in Section 3 indicates that recent models generally adopt newer base LLMs while retaining the CLIP ViT-L/14 vision encoder of Radford et al. (2021). We isolate the effect of each of these two base-model choices on fine-grained classification performance.

4.2.1 LLM CHOICE

Setup. Taking LLaVA-1.5-7B (Liu et al., 2023) as the baseline, we vary the LLM component while holding the vision encoder and pretraining configuration fixed within each comparison. We contrast the original Vicuna-7B-1.5 (Chiang et al., 2023) (used in LLaVA) against Qwen2-7B (Yang et al., 2024) (used in Qwen2-VL-7B (Wang et al., 2024b) and Molmo-7B-D (Deitke et al., 2024)), as well as against Llama2-7B (Touvron et al., 2023) (the base model of Vicuna) and Qwen2-7B-Instruct (the instruction-tuned variant of Qwen2-7B).

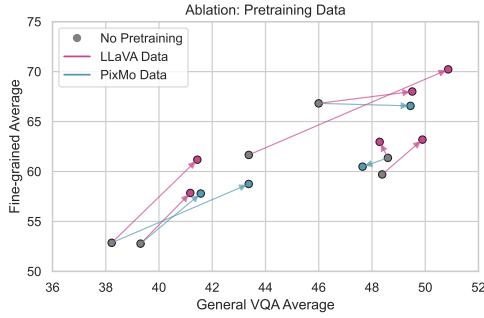


Figure 6: Pretraining-data ablation. Introducing either LLaVA (CC-3M) or Molmo (PixMo-Cap) pretraining data yields comparable gains: LLaVA data raises fine-grained and general accuracy by +4.0pp and +2.1pp respectively, and PixMo data raises them by +2.4pp and +2.5pp.

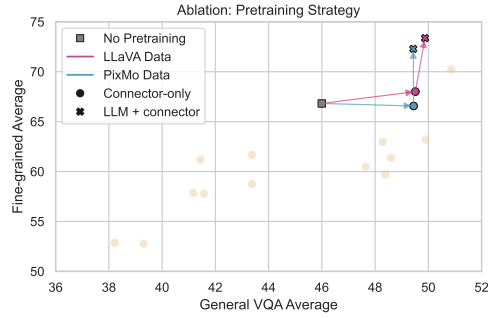


Figure 7: Training-method ablation: connector-only pretraining versus joint LLM-plus-connector pretraining. Unfreezing the LLM during pretraining yields a +5.5pp gain in fine-grained accuracy with no degradation of general VQA accuracy.

Findings. As reported in Figure 4, substituting Qwen2-7B for Vicuna-7B improves performance across all configurations, with mean gains of +7.5pp on the fine-grained benchmarks and +7.4pp on the general VQA benchmarks. The instruction-tuned Qwen2-7B-Instruct yields comparable gains (+7.5pp and +8.1pp, respectively). Llama2-7B, the non-instruction-tuned base of Vicuna, produces smaller gains, though the downstream consequences of omitting instruction tuning remain inconclusive. Critically, every trained model—including the lower-accuracy Vicuna-based VLMs—emits multiple-choice responses in the prescribed format essentially without exception. This controls for format compliance and localizes the fine-grained accuracy differences to the model’s knowledge of the presented options rather than its adherence to the output format.

Takeaway 1: Stronger language models consistently improve performance across both fine-grained classification and general VQA benchmarks.

4.2.2 VISION ENCODER CHOICE

Setup. Most evaluated VLMs, including the LLaVA series and Molmo, adopt OpenAI-CLIP ViT-L/14-336px (Radford et al., 2021) as their vision encoder. Qwen2-VL (Wang et al., 2024b), by contrast, adopts DFN-CLIP ViT-H/14-378px (Fang et al., 2023), which exhibits stronger zero-shot accuracy on object recognition tasks such as ImageNet. We evaluate both encoders across the LLM choices and pretraining configurations above.

Findings. Figure 5 shows that the effect of substituting DFN-CLIP H/14 for CLIP L/14 is conditional on whether connector pretraining precedes finetuning. In the absence of pretraining, DFN-CLIP degrades general VQA accuracy (-2.5pp) while modestly improving fine-grained accuracy (+2.8pp). Under connector pretraining, however, general VQA accuracy is preserved or marginally improved (+1.2pp) while fine-grained accuracy increases substantially (+4.5pp). Thus a stronger vision encoder confers a large fine-grained accuracy gain, but only once the encoder has been aligned to the LLM through pretraining.

Takeaway 2: Better vision encoders (like DFN-CLIP vs. CLIP) in VLMs improve fine-grained classification, but only when these encoders are properly integrated through pre-training before the finetuning stage.

4.3 TRAINING METHODS

Recent VLMs are predominantly trained in two stages (Liu et al., 2023; Wang et al., 2024b; Chen et al., 2024c). The first stage, pretraining, optimizes a captioning objective on a large-scale image-caption corpus to align the vision-encoder representations with the LLM embedding space. The

Table 1: **Summary of the principal ablation configurations, with existing models shown in bold for reference.** Substituting a stronger base LLM and vision encoder, together with pretraining the connector and LLM on sufficient data, each contribute to fine-grained performance, whereas instruction finetuning contributes comparatively little. The dependence differs for general VQA benchmarks, where the choice of LLM is the dominant factor.

Model/Ablation	Vision Encoder	LLM	Arch.	Pretraining Data	Finetuning Data	Fine-grained Classification	General VQA
LLaVA-1.5-7B	CLIP L/14	Vicuna	LLaVA	LLaVA	LLaVA	59.3	41.8
No pretraining	CLIP L/14	Vicuna	LLaVA	None	LLaVA	52.8	39.3
LLaVA reproduction	CLIP L/14	Vicuna	LLaVA	LLaVA	LLaVA	57.9 (+5.1)	41.2 (+1.9)
Qwen2 LLM	CLIP L/14	Qwen2	LLaVA	LLaVA	LLaVA	63.0 (+5.1)	48.3 (+7.1)
DFN-CLIP encoder	DFN H/14	Qwen2	LLaVA	LLaVA	LLaVA	68.0 (+5.0)	49.5 (+1.2)
Unfreeze LLM	DFN H/14	Qwen2	LLaVA	LLaVA	LLaVA	73.4 (+5.4)	49.9 (+0.4)
Pretrain on PixMo	CLIP L/14	Qwen2	LLaVA	PixMo	LLaVA	66.6	49.4
Molmo-7B-D	CLIP L/14	Qwen2	Molmo	PixMo	Molmo	68.4	58.0
FT Qwen2-VL base on LLaVA	DFN H/14	Qwen2	Qwen2-VL	Qwen2-VL	LLaVA	85.5	61.1
Qwen2-VL-7B	DFN H/14	Qwen2	Qwen2-VL	Qwen2-VL	Qwen2-VL	87.9 (+2.4)	62.4 (+1.3)

second stage, instruction tuning, adapts the VLM on curated multi-turn multimodal instruction-completion data to align its outputs with human intent for downstream applications. We now quantify the contribution of each of these two stages to the fine-grained capabilities of VLMs.

4.3.1 PRETRAINING

Setup. We address three questions concerning the pretraining stage:

- Is a pretraining stage necessary for LLaVA-architecture models whose base components are already pretrained on billion-scale data?
- How does connector-only pretraining compare against jointly unfreezing the LLM?
- To what extent does pretraining-data quality govern downstream performance?

We instantiate three pretraining configurations: LLaVA pretraining data (Liu et al., 2023) (a subset of CC-3M with web-scraped captions), Molmo pretraining data (Deitke et al., 2024) (PixMo-Cap, comprising high-quality human-annotated detailed captions), and no pretraining. In most configurations we update only the connector, following Liu et al. (2023). We additionally evaluate the schedule of Deitke et al. (2024), in which the connector is updated in isolation for the first 20% of steps and the LLM is jointly unfrozen for the remaining 80%. Every configuration is succeeded by one epoch of finetuning on the LLaVA finetuning data.

Findings. Figure 6 reports the pretraining ablation. Connector-only pretraining improves VLM performance in nearly all configurations, the sole exception being the Qwen2 with CLIP ViT-L/14 pairing. The gains are largest for Vicuna-based VLMs, which improve on average by +6.1pp in fine-grained classification and +3.1pp in general VQA.

Takeaway 3: Large-scale pretraining on image captioning datasets substantially improves fine-grained classification performance but has a more modest effect on general VLM benchmarks.

Unexpectedly, pretraining on low-quality web-scraped captions (LLaVA) and on highly detailed human annotations (PixMo) yields nearly indistinguishable outcomes: relative to LLaVA data, PixMo data induces a change of -1.6pp in fine-grained accuracy and +0.4pp in general VQA accuracy.

Takeaway 4: Pretraining data quality has a limited impact on overall model performance.

We hypothesize that connector-only pretraining constrains the model from exploiting the additional signal in higher-quality captions, because the LLM parameters are held frozen. To test this hypothesis, we jointly update the LLM and connector, preceded by a 20% connector-only warmup phase. Figure 7 shows that this schedule substantially improves fine-grained classification accuracy (+5.5pp) while leaving general VQA accuracy intact. The gain, however, is observed under both LLaVA and PixMo pretraining data, indicating that updating the LLM on the lower-quality captions does not degrade its language capabilities.

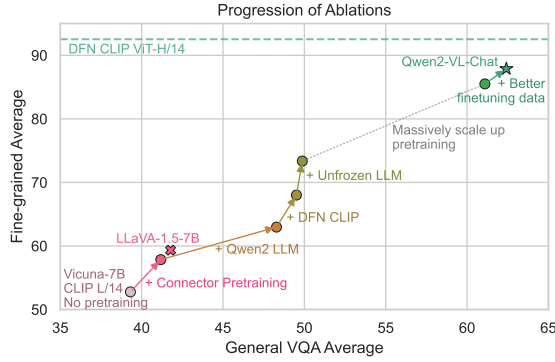


Figure 8: **Cumulative effect of the combined ablations.** Beginning from Vicuna-7B with CLIP ViT-L/14 and sequentially substituting the base LLM, the vision encoder, and the training configuration raises fine-grained classification accuracy from 52.8% to 73.4%, closing more than half of the 35.1pp gap to Qwen2-VL-Chat’s 87.9%. We attribute the residual gap primarily to the scale of Qwen2-VL’s pretraining corpus (1.4T tokens).

Takeaway 5: Pretraining both LLM and connector substantially enhances fine-grained benchmark performance while maintaining general VQA scores.

4.3.2 FINETUNING

Setup. Departing from the preceding protocol, we initialize from the pretrained Qwen2-VL-7B-Base model (Wang et al., 2024b) (the checkpoint from which Qwen2-VL-7B-Chat is trained) and finetune on LLaVA-Instruct data (Liu et al., 2023) under our earlier experimental protocol. We compare against Qwen2-VL-7B-Chat, which is instruction-finetuned on 2M in-house examples (Wang et al., 2024b). Holding the pretrained backbone fixed, this comparison isolates the effect of the finetuning dataset.

Findings. Finetuning exclusively on the LLaVA finetuning data marginally reduces both fine-grained and general benchmark performance. Fine-grained accuracy decreases by 2.4pp (from 87.9% to 85.5%) and general benchmark accuracy by 1.3pp (from 62.4% to 61.1%). As shown in Figure 8, this is the smallest contribution to fine-grained performance among all ablations, indicating that instruction finetuning plays a subordinate role in fine-grained knowledge acquisition relative to base-model selection and pretraining strategy.

Takeaway 6: The instruction finetuning stage has comparatively less impact on fine-grained classification performance than other factors.

4.4 SUMMARY OF ABLATIONS

Figure 8 and Table 1 synthesize the ablation study, decomposing the contribution of each component to closing the performance gap between the weakest and strongest VLMs on both fine-grained and general VQA tasks. Starting from the baseline LLaVA architecture (Liu et al., 2023) (Vicuna-7B (Chiang et al., 2023) with CLIP ViT-L/14 (Radford et al., 2021)) and applying the modifications sequentially raises fine-grained accuracy from 52.8% to 73.4% and general VQA accuracy from 39.3% to 49.9%. The decomposition shows that substituting DFN-CLIP (Fang et al., 2023) and unfreezing the LLM during pretraining disproportionately increase fine-grained capability relative to general capability, whereas substituting Qwen2-7B (Yang et al., 2024) for the LLM accounts for the majority of the increase in general VQA accuracy.

Even after these modifications, a residual 12-pp gap in fine-grained classification accuracy remains unaccounted for by our ablations (Table 1). We consider two candidate explanations for this residual: architectural differences and pretraining-data scale.

Although architecture may contribute, the observational results (Section 3.4) and the training ablations (Section 4.3) jointly implicate pretraining-data scale as the dominant factor. Our LLaVA (Liu

et al., 2023) and PixMo (Deitke et al., 2024) pretraining corpora are comparatively small—each fewer than 1M image-caption pairs, corresponding to approximately 200M and 400M tokens, respectively. By contrast, Wang et al. (2024b) report pretraining Qwen2-VL on 1.4T tokens, three to four orders of magnitude larger. This disparity in pretraining scale is the most plausible account of the residual gap, underscoring that large-scale pretraining is a critical prerequisite for VLMs with strong fine-grained classification capabilities.

5 LIMITATIONS & CONCLUSION

Our study is subject to practical limitations that delineate directions for future work. Under our computational budget, the ablations are restricted to pretraining corpora of <1M examples rather than the billion-scale (B) corpora used by some recent VLMs; the extent to which our conclusions extrapolate to that regime therefore remains open. Moreover, recent work proposes alternative pretraining strategies whose interaction with fine-grained visual understanding is not characterized here.

In summary, this work provides a systematic evaluation of state-of-the-art vision-language models (VLMs) on fine-grained classification benchmarks and establishes fine-grained visual classification as a critical yet underexamined dimension of VLM capability. Through a controlled analysis of individual model components and training procedures, we identify the design levers that most strongly govern fine-grained classification and vision-centric capability, thereby specifying concrete strategies for improving the reliability of VLMs in real-world settings that demand precise visual understanding.

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Vision Encoder	LLM	Arch	PT Data	FT Data	Updating	Classification	VLM
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	59.3	41.8
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	59.0	41.2
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	None	LLaVA	Connector	52.8	39.3
CLIP ViT-L/14@336	Llama2-7B	LLaVA	None	LLaVA	Connector	52.5	41.3
DFN ViT-H/14@378	Llama2-7B	LLaVA	None	LLaVA	Connector	56.2	40.0
DFN ViT-H/14@378	Vicuna-7B	LLaVA	None	LLaVA	Connector	52.9	38.2
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	None	LLaVA	Connector	61.4	48.6
DFN ViT-H/14@378	Qwen2-7B	LLaVA	None	LLaVA	Connector	66.8	46.0
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	59.7	48.4
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	61.7	43.4
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	LLaVA	Full	85.5	61.1
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	57.9	41.2
DFN ViT-H/14@378	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	61.2	41.4
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	63.0	48.3
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	68.0	49.5
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	63.2	49.9
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	70.2	50.9
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	Qwen2-VL	Full	87.9	62.4
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	57.8	41.6
DFN ViT-H/14@378	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	58.8	43.4
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	60.5	47.6
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	66.6	49.4
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Full	73.4	49.9
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Full	72.3	49.4

Table 2: **Per-configuration model ablation results (aggregated fine-grained classification and general VLM scores).**

Vision Encoder	LLM	Arch	PT Data	FT Data	Updating	MMBench	MMMU	MMStar
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	60.4	32.2	32.7
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	58.6	32.9	32.3
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	None	LLaVA	Connector	54.4	32.8	30.7
CLIP ViT-L/14@336	Llama2-7B	LLaVA	None	LLaVA	Connector	58.0	34.3	31.7
DFN ViT-H/14@378	Llama2-7B	LLaVA	None	LLaVA	Connector	53.9	34.6	31.5
DFN ViT-H/14@378	Vicuna-7B	LLaVA	None	LLaVA	Connector	51.3	33.2	30.1
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	None	LLaVA	Connector	63.1	41.8	40.9
DFN ViT-H/14@378	Qwen2-7B	LLaVA	None	LLaVA	Connector	61.3	36.2	40.5
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	63.2	41.4	40.5
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	55.7	38.3	36.1
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	LLaVA	Full	78.5	49.0	55.8
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	60.4	29.9	33.3
DFN ViT-H/14@378	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	57.4	34.2	32.7
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	64.4	41.3	39.1
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	64.9	41.3	42.3
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	66.5	41.3	41.9
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	68.0	41.6	43.0
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	Qwen2-VL	Full	78.9	49.9	58.4
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	57.8	32.6	34.3
DFN ViT-H/14@378	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	61.1	34.1	34.9
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	64.9	40.0	38.0
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	63.0	41.0	44.3
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Full	66.0	42.0	41.6
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Full	63.7	40.0	44.6

Table 3: **Per-configuration model ablation results, disaggregated by general VQA benchmark.**

A FULL RESULTS

We report the complete per-configuration ablation results in Table 2 (aggregated performance), Table 3 (per-benchmark general VQA), and Table 4 (per-benchmark fine-grained classification).

Vision Encoder	LLM	Arch	PT Data	FT Data	Updating	ImageNet	Flowers	Pets	Food
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	70.9	50.7	46.3	69.4
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	68.7	50.4	46.5	70.3
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	None	LLaVA	Connector	64.3	46.1	40.4	60.4
CLIP ViT-L/14@336	Llama2-7B	LLaVA	None	LLaVA	Connector	64.4	44.2	40.7	60.8
DFN ViT-H/14@378	Llama2-7B	LLaVA	None	LLaVA	Connector	65.4	50.5	46.2	62.8
DFN ViT-H/14@378	Vicuna-7B	LLaVA	None	LLaVA	Connector	62.3	48.7	39.2	61.3
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	None	LLaVA	Connector	70.6	44.1	55.6	75.2
DFN ViT-H/14@378	Qwen2-7B	LLaVA	None	LLaVA	Connector	74.1	50.8	62.3	80.1
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	71.4	39.6	53.1	74.7
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	72.0	46.0	54.1	74.5
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	LLaVA	Full	84.8	79.5	87.1	90.7
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	69.1	50.0	45.3	67.1
DFN ViT-H/14@378	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	72.7	53.2	48.5	70.4
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	71.5	46.9	56.6	76.9
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	76.1	54.1	59.1	82.7
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	74.5	45.8	55.1	77.4
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	78.5	52.3	65.4	84.7
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	Qwen2-VL	Full	85.9	82.3	91.0	92.3
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	69.2	47.4	46.7	67.8
DFN ViT-H/14@378	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	70.8	50.9	44.8	68.5
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	70.8	43.7	53.9	73.6
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	76.1	50.3	57.8	82.1
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Full	79.4	60.7	67.3	86.0
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Full	80.0	56.4	65.4	87.4

Table 4: **Per-configuration model ablation results, disaggregated by fine-grained classification benchmark.**

B EVALUATION

B.1 FINE-GRAINED CLASSIFICATION

Consistent with Section 3.2, we evaluate VLMs and base vision encoders on four traditional classification benchmarks:

ImageNet-1K (Deng et al., 2009): A foundational image classification dataset organized by the WordNet hierarchy, spanning a broad set of supercategories, each subdivided into multiple fine-grained subcategories. The test set comprises 50,000 images labeled across 1,000 classes.

The ImageNet/WordNet class hierarchy, however, contains many classes that are too visually distinct from the remaining classes to admit hard multiple-choice distractors (for example, the “person” supercategory contains only three classes). To address this—and to reduce the dataset’s excessive size—we use the human-annotated incorrect model predictions of Recht et al. (2019) to select images for which hard negative choices can be constructed. From an initial pool of 19,056 annotated examples, we retain 5,457 images spanning 928 of the original 1,000 classes. Because this curation leaves the retained classes imbalanced, we report mean-per-class accuracy for ImageNet.

Oxford Flowers-102 (Nilsback & Zisserman, 2008): A dataset of 102 flower species exhibiting high intra-class variation and high inter-class similarity. The test split contains 6,149 examples.

Oxford-IIIT Pet-37 (Parkhi et al., 2012): A collection of 37 pet categories with substantial variation in pose, illumination, and occlusion. The test split contains 3,669 examples.

Food-101 (Bossard et al., 2014): A dataset of 101 food categories exhibiting pronounced intra-class visual diversity. The test split contains 25,250 examples.

Question generation. For every fine-grained classification test set other than ImageNet (treated above), we generate hard negatives by zero-shot classification of the test examples with an OpenCLIP ViT-L/14 model trained on LAION (Schuhmann et al., 2022), following Geigle et al. (2024). We take the four highest-ranked incorrect predictions and shuffle them with the ground-truth label to form the answer set.

Prompt formatting. In the multiple-choice VQA format, the question is fixed to “What type of object is in this photo?” The answer choices are rendered as “A. Option 1\n B. Option 2\n C. Option

3\n D. Option 4\n E. Option 5”. Because different VLMs are trained with different multiple-choice prompts and formats, we adopt each model’s corresponding template from VLMEvalKit (Duan et al., 2024), e.g., the instruction “Answer with the option’s letter from the given choices directly.”

CLIP models are evaluated under the standard zero-shot procedure and hence use no question template; instead, each class name is passed to the text encoder under the template “a photo of a {classname}”.

B.2 GENERAL VQA

For the initial observational evaluation in Section 3, we use the scores reported directly on the OpenVLM Leaderboard (Duan et al., 2024). Its default metric is the mean score over eight diverse VLM benchmarks: MMBench (Liu et al., 2024b), MMStar (Chen et al., 2024b), MMMU (Val) (Yue et al., 2024), MathVista (Lu et al., 2024), OCRBench (Liu et al., 2024c), AI2D (Kembhavi et al., 2016), HallusionBench (Guan et al., 2024), and MMVet (Yu et al., 2024). This aggregate score is used only in Figure 2.

For all subsequent evaluations—of both existing models and the models we train in Section 4—we restrict to a subset of multiple-choice VLM benchmarks to enable a more direct and controlled comparison. Specifically, we evaluate on three benchmarks:

MMBench (Liu et al., 2024b): A dataset of 2,948 examples probing a broad range of VLM capabilities partitioned into perception and reasoning. Although most examples target coarse-grained perception and visual reasoning, a subset targets “fine-grained perception”; these categories (e.g., action recognition, attribute recognition, and OCR) nonetheless lie in a domain distinct from the fine-grained classification datasets we consider.

MMMU (Val) (Yue et al., 2024): A large-scale multimodal dataset spanning a wide distribution of domains and applications. Its test set contains 10,500 examples, but performance is conventionally reported on the 900-question validation split. MMMU probes domain knowledge grounded in visual perception across art, science, engineering, medicine, business, and the humanities.

MMStar (Chen et al., 2024b): A dataset of 1,500 human-selected questions chosen to cover a diverse set of tasks across both domains and low-level model capabilities.

Evaluation procedure. For the general VQA benchmarks, we defer entirely to the VLMEvalKit evaluation code (Duan et al., 2024). Existing models are scored with their predefined evaluation procedures; for our own trained models we apply the LLaVA evaluation code, as we finetune exclusively on LLaVA-Instruct data.

C TRAINING

All experiments are implemented on the LLaVA-1.5 training codebase (Liu et al., 2023), and, except where stated otherwise, follow the LLaVA architecture and training procedure.

Pretraining. Pretraining consists of one epoch on either the LLaVA pretraining data (CC-3M images and captions) or the Molmo pretraining data (PixMo-Cap). For PixMo-Cap we use only the “caption” field and discard the raw audio transcript. Pretraining uses a batch size of 256. In most runs we update only the randomly initialized MLP connector, at a learning rate of $1e-3$. In the pretraining-strategy ablation, we first update the connector alone as a warmup for the initial 20% of steps, then unfreeze and jointly update the LLM at a learning rate of $2e-5$ for the remaining 80% of steps.

Finetuning. Finetuning consists of one epoch on the LLaVA-1.5 finetuning data, which comprises LLaVA-Instruct together with VQA training data drawn from several datasets to support standardized benchmarking. During finetuning we set the LLM learning rate to $2e-5$ with a batch size of 128. The vision encoder is held frozen throughout both training phases.

For the Qwen2-VL-architecture ablation, we initialize from Qwen2-VL-7B Base and apply the same procedure, updating only the LLM.

Compute. All models are trained on 4 nodes, each with 4 A100 (40GB) GPUs. Training completes within approximately 20 hours, varying with the size of the CLIP encoder.

LLaVA vs. PixMo data. The main text contrasts the caption quality of the LLaVA and PixMo pretraining corpora. The LLaVA pretraining data derives from CC-3M, consisting of web-scraped captions that are typically of low quality and relevance. The PixMo-Cap data, in contrast, is constructed by having human annotators record a 60–90 second audio description of an image, which is then transcribed into a coherent image caption (Deitke et al., 2024). Quantitatively, LLaVA captions average 9.8 words, whereas PixMo captions average 169 words.

LLaVA caption example: “front panel bracket cover for suzuki”

PixMo caption example: “The poster from the TV show ”Law and Order” features a past cast ensemble set in an interrogation room. On the left is a Hispanic man, identified as Detective Nick Amaro, portrayed by Danny Pino. He has dark hair, olive skin, and is dressed in a gray jacket, blue shirt, and black tie. Standing beside him is a blonde woman with shoulder-length hair, who is identified as Detective Amanda Rollins, played by Kelli Giddish. She is ... The backdrop is an interrogation room, with a concrete wall and a portion of a two-way mirror visible, adding to the procedural drama’s atmospheric setting.”